

From SBTG Gradient Readouts to Compatibility-Aware Path Responses

A technical record of the models, synthetic systems, metrics, failures, repairs, and empirical results

Repository synthesis for `new_sbtg_neuro`

Evidence reviewed through 2 September 2026

Abstract

This report reconstructs the full methodological arc of the repository. The project began with score-based temporal graph (SBTG) statistics intended to reveal directed, lagged, and distributional neural relationships. Repeated improvements changed the architecture, score objective, derivative readout, regularization, scale, conditioning scheme, and inference wrapper. Those changes produced real but narrow advances: nonlinear edge-local SBTG improved over the original feature-bilinear window model; a residual-score temporal map made clean mean-channel recovery competitive with ridge VAR; and conditional variance improved held-out neural prediction. They did not produce a robust receptor-specific, anatomical, or physical-delay claim.

The decisive synthetic lesson was that a conditional-mean model is excellent when the truth is encoded in the mean, but mean-only summaries are mathematically blind when histories alter covariance, reliability, skewness, mixture occupancy, or higher-order path shape while leaving the mean fixed. Flexible conditional laws can predict those distributions better, yet derivatives of the learned law can remain noisy or wrong. Averaging conditional-mean effects over heterogeneous histories can also cancel signal and amplify finite-sample instability.

The current program therefore changes the scientific object. It defines a finite, supported, compatibility-aware contrast between complete model-generated histories, rather than an arbitrary coordinate edit. Direct weighting and sequential Monte Carlo estimate the corresponding future laws; progressive bridge SMC improves finite-particle approximation for selective queries. This is a model-relative predictive response, not an identified physical intervention. Support, cross-model disagreement, quiet-time controls, and learned-model error remain mandatory because a numerically healthy sampler cannot repair a wrong predictive model.

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1 The result in one page

Stage	What worked	What failed	Current interpretation
Original SBTG	Moderate mean-like edge localization; edge-local nonlinear terms added small AUROC gains	Weak variance/gating specificity; feature-bilinear SBTG often near chance at 50–100 nodes	Reduced-form screen, not a conditional density or mechanism estimator
Conditional-score repair	Corrected conditional-score math; residual temporal map reached AUROC 0.720 ± 0.045 without measurement noise	Unconstrained score Jacobian stayed near 0.50; measurement noise reduced the repaired result to 0.578 ± 0.014	The useful signal was mostly the conditional mean map
Matched conditional laws	Gaussian NLL led mean and variance-field recovery; MDN led matched-tail prediction	Tail derivatives and synaptic gating remained worse than zero baselines	Predictive fit and derivative/mechanism recovery are different endpoints
Finite law contrasts	Location and low-rank mixture witnesses beat zero; later ORTH/typed repairs recovered selected covariance geometry	Initial covariance/skew adapters failed even with oracle samples; unrestricted full-feature recovery remained poor	Use supported finite contrasts and declared response channels
Neural application	Conditional variance improved leave-one-worm-out NLL by +0.364 on 28 worms; aggregation improved reproducibility	Original peptide lag signal attenuated from +3.35 to -0.41 ; kinetics, gating, shrinkage, and receptor localization failed	Strong predictive-distribution result, negative mechanistic result
Compatibility response	Progressive SMC reduced finite-particle MSE to 0.0110 and raised valid rate to 0.911	Sixteen synthetic rows were numerically stable but scientifically wrong because the learned model was wrong	Current estimator of choice only with support/model diagnostics
Current atlases	Corrected 54-neuron and historical 80-class artifacts are complete and audited	Corrected atlas had zero rows passing both sampler and quiet-time controls; historical 80-class lineage is invalid for prospective evidence	Observational hypothesis generator, not a causal or anatomical graph

Interpretation. The central shift is from asking whether a local score derivative resembles an edge to asking how the fitted future law changes between two supported classes of complete histories. The first is an ill-conditioned derivative/localization problem; the second is a declared finite-law contrast with explicit overlap and Monte Carlo diagnostics.

2 Claim axes and terminology

The same numerical object must not be promoted across scientific claim levels. We use the following hierarchy throughout.

Axis	Question	Examples	Maximum defensible wording
P1	Does the model predict held-out observations?	NLL, CRPS, energy score, coverage, RMSE	Predictive accuracy or calibration
D1/D2	Does the conditional law or path law change?	Variance, tail, W1, finite witness, path contrast	Observed conditional/distributional response
M1/M2	Does a fitted readout align with a declared mean or dispersion field?	Mean Jacobian, log-variance field, typed covariance derivative	Model-relative or synthetic mechanism-matched recovery
L1	Is a represented latent state recovered?	Modulator concentration, clearance, occupancy	Latent alignment under the specified model
C1	Does a physical intervention have a controlled effect?	Shared-noise knockout, randomized perturbation	Causal only under explicit identification assumptions

Table 2: Claim axes used to prevent predictive success from being mislabeled as mechanism or causality.

Real neural results in this document are observational and model-relative unless explicitly stated. Anatomical connectomes, receptor maps, and perturbational atlases are external correspondence references, not ground truth for the fitted response matrices.

3 Mathematical objects and why the original route became unstable

3.1 Conditional law, mean, and variability

Let $H \in \mathcal{H}$ be a neural/stimulus history and $Y \in \mathbb{R}^d$ a future response. The fundamental predictive object is

$$P_h(A) = \Pr(Y \in A \mid H = h), \quad p_h(y) = p(y \mid h). \quad (1)$$

The conditional mean and covariance are

$$m(h) = \mathbb{E}[Y \mid H = h], \quad \Sigma(h) = \mathbb{E}[(Y - m(h))(Y - m(h))^\top \mid H = h]. \quad (2)$$

A mean model estimates $m(h)$. It is optimal for squared-error prediction, and it recovers mean-driven synthetic wiring well. But if P_{h_+} and P_{h_-} have equal means and differ only in covariance, skew, mixture weights, reliability, or path topology, then $m(h_+) - m(h_-) = 0$ even when the laws are different.

Dataset averaging creates a second difficulty. A reported average response

$$\bar{\Delta} = \frac{1}{n} \sum_{i=1}^n \{m(h_i^+) - m(h_i^-)\} \quad (3)$$

can be noisy or nearly zero when local effects vary in sign across histories. More rows do not necessarily create more independent information when rows share worms, events, or overlapping time windows.

3.2 History score, SBTG operator, and derivative amplification

For a history direction v and perturbation $T_{\epsilon,v}h$, define the infinitesimal history witness

$$\ell_v(y; h) = D_v \log p_h(y) = \left. \frac{\partial}{\partial \epsilon} \log p_{T_{\epsilon,v}h}(y) \right|_{\epsilon=0}. \quad (4)$$

It satisfies $\mathbb{E}_h[\ell_v(Y; h)] = 0$ and, for a declared response feature ϕ ,

$$D_v \mathbb{E}_h[\phi(Y)] = \mathbb{E}_h[\phi(Y) \ell_v(Y; h)]. \quad (5)$$

The conditional outcome score is $s_y(y; h) = \nabla_y \log p_h(y)$. The ideal SBTG mixed operator is

$$K_v(y; h) = D_v s_y(y; h) = \nabla_y \ell_v(y; h). \quad (6)$$

Thus SBTG sees a response-space gradient of the history witness. Recovering ℓ_v from K_v requires integration, centering, support, and boundary conditions. A cross-moment of a learned joint-state score is not automatically a conditional-law derivative, a physical field, or a causal graph.

Hyvärinen score matching minimizes, up to constants,

$$\mathcal{L}_{\text{SM}}(\theta) = \mathbb{E} \left[\frac{1}{2} \|s_\theta(Y, H)\|_2^2 + \nabla_Y \cdot s_\theta(Y, H) \right]. \quad (7)$$

Denoising score matching instead regresses the score of a Gaussian-corrupted law. A small response-score error does not imply a small history derivative error:

$$s_{\theta, n} \rightarrow s_y \not\Rightarrow D_v s_{\theta, n} \rightarrow D_v s_y. \quad (8)$$

This was observed directly: autodifferentiation and finite differences agreed to relative error about 1.4×10^{-4} , while graph AUROC stayed near chance. The code was differentiating the learned model correctly; the learned derivative was the wrong scientific statistic.

3.3 Finite supported contrast

For $h_\pm = h \pm \delta v$, define

$$M_\delta = \frac{1}{2}(P_{h_+} + P_{h_-}), \quad W_\delta(y) = \frac{1}{\delta} \tanh \left(\frac{\log p_{h_+}(y) - \log p_{h_-}(y)}{2} \right). \quad (9)$$

The bounded witness W_δ represents the central signed contrast at a declared scale. More generally, if $\nu_c = \sum_r c_r P_{h_r}$, $\sum_r c_r = 0$, and $M_c = C^{-1} \sum_r |c_r| P_{h_r}$ with $C = \sum_r |c_r|$, then

$$W_c = \frac{d\nu_c}{dM_c}, \quad |W_c| \leq C, \quad \sum_r c_r \mathbb{E}_{h_r}[\phi(Y)] = \mathbb{E}_{M_c}[\phi(Y)W_c(Y)]. \quad (10)$$

This target can be estimated by normalized densities, density ratios, balanced classifiers, projected moments, or samples. It avoids differentiating a flexible model with respect to history, but it still requires overlap and a response feature capable of seeing the target change.

3.4 Compatibility-aware repaired paths

The later neural work uses complete generated path prefixes. For prediction cut c , source j , lag ℓ , and source-window width w , let

$$A_j(Z) = \frac{1}{w} \sum_{r=c-\ell-w+1}^{c-\ell} Z_{rj}. \quad (11)$$

Training-fold quartiles $q_{j,.25}, q_{j,.75}$ define soft source potentials

$$G_j^a(Z) = \exp \left[-\frac{1}{2} \left(\frac{A_j(Z) - q_{j,a}}{b_j} \right)^2 \right], \quad b_j = \max(0.25 \text{IQR}_j, 0.10). \quad (12)$$

An anchor potential $G_{\text{anc},j}(Z) = \exp[-\lambda C_j(Z, Y)]$ keeps non-source population context near the factual episode. The repaired path law is

$$Q_j^a(dZ) \propto P_\theta(dZ \mid H_{c-w-\ell}, U) G_{\text{anc},j}(Z) G_j^a(Z). \quad (13)$$

For future target feature ϕ_k , the saved response per achieved source unit is

$$M_{kj}^{(\ell)} = \frac{\mathbb{E}_{Q_j^{.75}}[\phi_k] - \mathbb{E}_{Q_j^{.25}}[\phi_k]}{\max(|\hat{A}_{j,\text{high}} - \hat{A}_{j,\text{low}}|, 0.10)}. \quad (14)$$

This resembles backtracking counterfactual reasoning because the past is revised while downstream dynamics remain generated by the same model. It is deliberately called a compatibility-aware, model-relative response rather than a structural counterfactual.

4 Metrics and evaluation rules

Metric	Formula/implementation	What it measures	Critical caveat
NLL	$-n^{-1} \sum_i \log q_\theta(y_i h_i)$	Proper score for a normalized predictive density; lower is better	Not available for unnormalized score models
RMSE/NRMSE	$\sqrt{n^{-1} \sum (\hat{t} - t)^2}$, divided by target scale	Point or witness error; zero-estimator NRMSE is near 1	Depends on target normalization and panel
nISE	$\int \ \hat{f} - f\ ^2 / \int \ f\ ^2$ over occupied synthetic states	Field recovery against known truth; zero field equals 1	Synthetic/mechanism-specific only
Tail RMSE	RMSE of fitted versus oracle tail probabilities	Predictive tail calibration	Does not test the derivative of tail probability
AUROC/AUPRC	Ranking of positive versus negative reference edges	Support/localization correspondence	Not coefficient calibration or causality
AP excess	AUPRC minus positive-edge prevalence	Graph recovery adjusted for class prevalence	Still observational under hidden confounding
Energy/CRPS	Proper sample scores using predictive draws	Full distributional prediction without density evaluation	Does not certify derivative quality
Coverage	Fraction of oracle coefficients covered by intervals	Calibration of stated uncertainty	Requires correct inferential unit and multiplicity policy
Hit within 5s	Scan maximum lies within 5 seconds of the known boundary	Localization	Can be high while calibrated detection is zero
Detected within 5s	Localized and passes the calibrated p/q threshold	Detection plus localization	Calibration scheme must preserve time/subject dependence
ESS fraction	$(\sum_i w_i)^2 / (N \sum_i w_i^2)$	Particle-weight degeneracy	Healthy ESS validates sampling of a model, not the model
Response MSE	Error versus two independent 4,096-particle references for the same fitted law	Finite-particle estimator accuracy	Not error versus biological truth
Worm sign-flip/max- T	Joint sign flips of worm-level vectors	Frozen-model animal-level evidence with family correction	Overlapping training folds remain coupled

5 Synthetic data actually used

5.1 Matched mechanistic objective panel

The most direct objective comparison used one mechanistic simulator, identical splits, horizon, and DGP seeds across ten methods. Each scenario had 10 trajectories, 10 neurons, 3 hidden modulators, 1 history lag, one-step horizon, 20% validation and 20% test fractions, 64 predictive samples, and complete-state observations. DGP seeds were 8–12. All 250 method-by-mechanism-by-seed cases completed.

Scenario	Strength	Steps/burn-in	Ligand process	Changed part of one-step law
Null	0.45	1000/1500	amplitude 2.5; rate 0.08 Hz	No active neuromodulatory channel
Additive mean	0.35	1000/1500	same	Receptor occupancy adds to conditional mean
Synaptic gain	1.20	1000/1500	same	Occupancy multiplicatively gates recurrent weights
Stochastic dispersion	1.50	1000/1500	same	Occupancy changes log innovation variance
Matched tail	2.50	1000/1500	same	Occupancy changes mixture probability while normalization preserves mean/variance scale

Table 4: Frozen matched-panel data-generating conditions.

The simulator used positive release, persistent concentration, Hill receptor occupancy, recurrent dynamics, additive drive, log-variance modulation, and a two-scale Gaussian mixture for the matched-tail mechanism. Counterfactual arms shared exogenous noise where controlled operations were evaluated.

5.2 History-tangent generators G1–G8

Histories were sampled independently from $N(0, I)$ except G7 and G8, whose support/path processes were explicit. Exact log densities and automatic-differentiation oracles were implemented in double precision, separately from trainable models.

DGP	Default size	Exact construction	Purpose
G1	$q = 8, d_y = 4$	$Y = \tanh(HW_1^\top)W_2^\top + \epsilon$ with rotated Gaussian covariance and SDs 0.5–1.0; hidden width 16	Nonlinear location-only law
G2	$q = 8, d_y = 4$	Rotated Gaussian with eigenvariance $\exp\{b + 0.6 \tanh(AH/\sqrt{q})\}$; mean coefficient $\alpha_m \in \{0, 0.3\}$	Covariance-only or covariance-plus-mean change
G3	$q = 8, d_y = 1$	Four Gaussian components at $(-3, -1, 1, 2)$, SD 0.25; weights $0.25 + 0.12 \tanh(H^\top v)(-0.2, 2/3, -1, 8/15)$	Skew change with exactly fixed mean and variance
G4	$q = 8, d_y = 8$	Four isotropic Gaussian components; softmax logits BH , rank 2; mode separation 4σ	Low-rank multimodal occupancy change
G5	$q = 8, d_y = 32$ (focused $d_y = 64$)	Eight components in a 2-D (focused 4-D) latent manifold embedded in response space; $\sigma_z = 0.5$, $\sigma_{obs} = 0.1$	Difficult high-dimensional generation geometry
G6	$q = 8, d_y = 8$	Smooth location plus $(0.5/8) \sin(8H_1)u$ and unit Gaussian noise	Small density perturbation with order-one high-frequency derivative
G7	$q = 32, d_y = 8$	$H = ZB^\top + 0.1\epsilon$, intrinsic dimension 4, with G2 or G4 conditional law on Z	Near-manifold history support and off-support directions
G8	$q = 25, d_y = 48$	Six history steps and 12 future steps over four channels; 16 signed motif components, motif scale 0.45, noise SD 0.08, correlated scale 0.06; a control changes frame mixture with sensitivity 1.2 while mean/covariance stay fixed	Pure higher-order whole-path shape/order effect

5.3 Distributional-versus-mean circuit battery

The comparable battery used $N = 40$ latent nodes, a randomly observed $M = 12$, $L = 3$ lags, $T = 20,000$ unit-stride steps, and sparse off-diagonal operators with Bernoulli density 0.05. Per-lag spectral radii were $0.6/\sqrt{L}$ for mean

operators, $1.0/\sqrt{L}$ for variance operators, and $3.0/\sqrt{L}$ for covariance routing. Each trajectory was clipped to $[-12, 12]$; the observed $[H | Y]$ block was column-standardized. The primary metric was off-diagonal lag-union graph AUROC against the known observed-to-observed parent support. Held-out mean R^2 diagnosed whether the conditional mean carried usable signal.

Mechanism	Exact one-step construction	Edge-bearing object	Conditions
Linear	$X_{t+1} = \sum_{\ell=0}^{L-1} X_{t-1-\ell} A_{\ell}^{\top} + 0.5\eta$, $\eta \sim N(0, I)$	Conditional mean	Clean
Tanh	$X_{t+1} = \tanh(\sum_{\ell} X_{t-1-\ell} A_{\ell}^{\top}) + 0.4\eta$	Conditional mean	Clean
Gated	$X_{t+1} = (0.5 + 0.5 \tanh X_{t,g}) \sum_{\ell} X_{t-1-\ell} A_{\ell}^{\top} + 0.4\eta$	Conditional mean with state gate	Clean
Branch	$X_{t+1} = b \odot \sum_{\ell} X_{t-1-\ell} A_{\ell}^{\top} + \eta$, with coordinatewise random sign $b \in \{-1, +1\}^N$	Marginal variance; $\mathbb{E}[Y H] = 0$	Clean; observed noise $0.4N(0, I)$
Variance gate	$X_{t+1} = s(H) \odot \eta$, $s(H) = 0.3[1 + 2 \tanh \sum_{\ell} X_{t-1-\ell} V_{\ell}^{\top}]$	Marginal conditional variance/reliability	Clean; observed noise $0.4N(0, I)$
Cross covariance	$X_{t+1,j} = \sqrt{1 - a_j} \epsilon_j + \sqrt{a_j} \xi_{c(j)}$, $a_j = \text{sigmoid}(2.5[C \cdot H]_j)$	Off-diagonal covariance; every marginal has mean 0 and variance 1	Clean; observed noise $0.4N(0, I)$

For the covariance row, $c(j)$ assigns a node to one of three clusters at $N = 40$; ϵ_j is private unit Gaussian noise and $\xi_{c(j)}$ is a cluster-shared unit Gaussian innovation. Thus $\text{Cov}(Y_a, Y_b | H) = \sqrt{a_a a_b} \mathbf{1}\{c(a) = c(b)\}$ while every per-target conditional marginal remains $N(0, 1)$. The first per-target study used three seeds and seven mean/variance conditions; the later no-free-lunch adjudication used ten seeds and added the joint covariance rung.

6 Every implemented method and what it technically estimated

The tables below list distinct implemented estimators or materially different readouts. Hyperparameter variants of the same mathematical estimator are grouped only when their target and conclusion were identical.

6.1 Classical dynamics and normalized predictive laws

Method	Technical implementation	Learned quantity	Empirical role
Constant Gaussian	History-independent mean/variance reference	Marginal Gaussian law	Null/simple prediction baseline
Ridge VAR	Ridge least squares on lagged neural history	Linear conditional mean and lag coefficients	Strongest routine mean baseline
VAR-LASSO	L1-penalized lag regression	Sparse conditional-mean graph	Strong clean gated baseline
Conditional Granger ridge	Restricted versus full ridge regression by source history	Predictive contribution of a source	Strong graph baseline; observational
Sparse transition	Thresholded/penalized transition matrix	Sparse one-step mean dynamics	Strong matched mean recovery
Polynomial SINDy-LASSO	Sparse library regression over polynomial state features	Nonlinear conditional mean	Competitive but weaker than oracle library
OracleLibrary SINDy	SINDy using the generator-matched nonlinear basis	Best-case sparse mean dynamics	Strongest clean synthetic comparator
PCMCI partial correlation	Conditional-independence graph from lagged partial correlations	Observed lag support	Degrades under hidden drive

Method	Technical implementation	Learned quantity	Empirical role
VAR-LiNGAM	Linear non-Gaussian acyclic mixing applied to VAR residual structure	Directed linear support under non-Gaussian assumptions	Best tested strong-confounder graph baseline, still not identified
Heteroskedastic ridge	Ridge mean plus regression for conditional scale	Gaussian mean/variance law	Cheap distributional baseline
Variance regression	Fit a mean, then regress squared residuals on H and differentiate the fitted variance	Marginal conditional-variance support	Matched or beat the residual-score variance head
Cross-product regression	Regress residual products $R_a R_b$ on H and differentiate/partner-average	Off-diagonal conditional covariance support	Matched or beat joint score/transport readers
Penalized Gaussian NLL	Jointly optimized conditional mean and log variance with ridge penalty	Normalized Gaussian law and analytic fields	Best matched mean/dispersion-field method
Student- t ridge	Linear location/scale with normalized Student- t likelihood; df tuned in the matched panel	Heavy-tailed conditional law	Best cheap robust tail baseline
Gaussian MLP NLL	Two-layer 64-unit MLP heads for mean/log variance; early stopping by validation likelihood	Nonlinear normalized Gaussian law	More capacity did not beat simple Gaussian NLL
Mixture density network	Neural mixture weights, locations, and scales; $K = 3$ matched panel, $K = 4$ later OOD work	Multimodal normalized law and sampler	Best matched-tail prediction
Autoregressive MDN	Coordinate-wise mixture conditionals with ancestral sampling	Normalized vector law and direct history gradient	Tangent error near/above zero baseline
Causal autoregressive Transformer	Teacher-forced continuous mixture likelihood under three fixed coordinate orders	Normalized autoregressive law	Good location tangents; no hard-channel validity
Affine conditional flow	Invertible conditional affine transport with exact density	Normalized flexible law	No tangent advantage in pooled benchmark
Neural spline flow (NSF-6/10)	Conditional rational-quadratic spline flow with 6 or 10 transforms	Reusable normalized conditional law	E10 flexible-law comparator; no universal winner

6.2 Score, diffusion, and SBTG variants

Method	Technical implementation	Readout/target	Result
Quadratic SID, Hyvärinen	Quadratic response-score basis; closed-form penalized score-matching solve; Gaussian reconstruction	Mean, log-variance, and tail-field derivatives	Viable in matched panels, weaker than Gaussian NLL
Quadratic SID, DSM	Same quadratic family trained on Gaussian-corrupted outcomes with 4 corruptions; $\sigma = 0.75, 1.0$	Corrupted-law score and reconstructed fields	Regularized some fits; dispersion remained at/worse than zero
Conditional score MLP	Future-only neural score fit by Gaussian or Student- t DSM	$\nabla_y \log p_\sigma(y h)$ and mixed history Jacobian	Score reconstruction high while edge AUROC near chance

Method	Technical implementation	Readout/target	Result
Gaussian MLP DSM	Gaussian parameter heads optimized through DSM; $\sigma = 0.25, 0.50$	Corrupted Gaussian score and analytic parameter readouts	Small within-architecture dispersion gain, no overall win
Legacy VE diffusion	Variance-exploding denoising score network and reverse sampler	Noisy response score, samples, reconstructed history tangent	Pooled tangent median NRMSE 1.022
EDM diffusion	EDM-preconditioned score network; deterministic Heun sampler	Same with improved sampling parameterization	Pooled tangent median NRMSE 1.743; G5 generation not better
Gaussian-anchored diffusion	Diffusion score anchored to a Gaussian source/reference	Stabilized noisy-law score	Pooled tangent median NRMSE 2.692
Gaussian DSM baseline	Direct Gaussian denoising score network without full diffusion sampling	Corrupted response score	Pooled tangent median NRMSE 1.260
Stochastic interpolant / flow matching	Conditional drift regression along stochastic interpolation; Euler generation	Transport drift and samples	Stable on G1, nearly zero G8 quartic slope
Bounded energy model	Conditional bounded energy with score/autodiff readout	Unnormalized law and history tangent	Pooled tangent median NRMSE 1.181
SBTG linear	Linear joint score over consecutive standardized states	Score cross-moments by source/target	Often matched or beat feature-bilinear SBTG in frozen mechanism panel
SBTG FeatureBilinear	Joint-window DSM with within-time features and bilinear cross-time terms	Edge energy from score-product/cross-block terms	Original default; moderate mean-like, weak variance/gating specificity
SBTG NonlinearEdge	Edge-local nonlinear basis for each candidate cross-time edge	Edge-local window energy	Added roughly 0.061 AUROC at 50 nodes
SBTG NonlinearEdgeSparse	Edge-local nonlinear basis plus group sparsity	Sparse nonlinear edge energy	Best window extension; still far below VAR
SBTG HybridEdgeSparse	Feature-bilinear and edge-local nonlinear terms with sparsity	Hybrid window energy	Similar small gain, no overall win
SBTG ConditionalPairDSM	Conditional pair score trained on history/future partitions	Conditional-pair window statistic	Some tuned cells improved, still below ridge
Direct conditional-score Jacobian	Train $s_\theta(y H, \sigma)$ and differentiate score w.r.t. each lagged history coordinate	Corrected score-history Jacobian	Mean parents recoverable; variance parents near chance
SmoothGrad, finite-difference, jacrev readouts	Average perturbed gradients; central finite differences; vectorized Jacobian	Robustness of the same learned derivative	Agreed numerically; no material AUROC improvement
Jacobian-consistency/Sobolev penalties	Penalize disagreement or oracle/local derivative mismatch during training	Smoother score Jacobian	Did not fix graph readout
Residual-score temporal Jacobian	Constrain $s_\theta = -(y - \mu_\theta(H))/(v_\theta(H) + \sigma^2)$ and score edges with $\partial\mu_\theta/\partial H$	Conditional-mean temporal map	Largest SBTG repair; competitive with ridge only in clean mean channel

Method	Technical implementation	Readout/target	Result
Residual variance / Hessian DSM	Differentiate predicted log variance or score curvature with respect to history	Reliability/second-order routing	Only robust distributional win in mean-blind battery; Hessian itself never won
Residual covariance MLP	Residualize Y , regress all $R_a R_b$ on H , differentiate, then average over pairs incident to each target	Off-diagonal covariance routing	Recovered the joint rung; linear cross-product regression was stronger

6.3 Finite, orthogonal, path, and inference methods

Method	Technical implementation	Target	Result
Point-history tangent	Autodifferentiate $\log q(y h)$ or a response score in direction v	Infinitesimal $D_v \log p_h$	Optimization passed; hard-channel readiness failed
Central model log ratio	Evaluate $\log q(y h + \delta v) - \log q(y h - \delta v)$	Finite law contrast	Useful when density normalized and histories supported
Joint/product ratio critic	Binary classifier separating joint history-response pairs from product pairs	Interaction ratio without history-only term	Best pooled legacy tangent median, but not a full history witness
Balanced signed classifier	Classify samples from P_{h+} versus P_{h-} ; held-out affine calibration	Bounded finite witness W_δ	Good G1 and G4; oracle adapter failed G2/G3
Signed Riesz dictionary	Solve regularized moment equations over linear/quadratic/cross/cubic/tail features	Projection of finite witness	Dictionary-limited; original hard-channel results worse than zero
PLUG	Plug nuisance estimates into a feature derivative functional	Point estimate of bounded-feature derivative	Similar point error in regular cells; naive intervals undercovered
RIESZ-only	Estimate history-side Riesz representer without orthogonal correction	Feature derivative functional	Useful diagnostic; nuisance-sensitive
ORTH	Cross-fit feature regression and Riesz representer; add orthogonal residual correction	Average derivative of declared predictive features	Best calibrated framework after target-specific repairs
Density-score Riesz	Fit BIC-selected history density/GMM and differentiate log density	Improved history representer for tails/gaps	Removed E6 fixed-sieve bias at large n
Characteristic features/inversion	Regress complex exponentials $e^{it^T Y}$; invert to typed covariance target	Moment-blind/full-law feature derivatives	Works only with resolving frequencies and small inversion ridge
Direct covariance/low-rank head	Estimate covariance derivative directly and project to effective rank	Typed covariance geometry	Strong E9 scaling; full feature tensor remained hard

Method	Technical implementation	Target	Result
Latent neuromodulated SSM	Sequential likelihood for release, concentration, Hill occupancy, recurrent state, and emission; explicit knockout mask	Represented latent/operation response	Higher claim ceiling, weak current recovery
Conditional Brownian bridge	Heteroskedastic endpoint regression plus Brownian reference bridge	Reference-relative path law	Useful D2 baseline; reference diffusion dominated
Schrödinger/transport bridge	Learn a conditional path transport and read the history dependence of transported samples	Joint conditional-law response	Recovered covariance routing, but did not beat matched regression
Direct repaired-path importance	Generate natural path bank and self-normalize low/high compatibility weights	Compatibility-aware future-law contrast	Fast; collapses for rare source states
Terminal SMC	Sequential proposal; apply source potential at window end; defer source resampling to cut	Same repaired law	Lower MSE than direct, but terminal collapse persists
Progressive bridge SMC	Branch proposals, partial-window look-ahead, adaptive tempering to ESS target, resample, release	Same repaired law with lower particle error	Best sealed finite-particle result
Temporal-cut SMC	Apply full lagged source constraint when its window closes; ESS-resample through gap	Earlier constraint placement under same general law	Explicit-lag sensitivity; early ancestry collapse risk
Changepoint scans	CUSUM, Hotelling, Chow, variance/reliability, MMD, energy, PELT, binary segmentation, score-product scans	Whether/where a declared statistic changes	Useful only after static channel validation

7 Experiment results: the SBTG repair sequence

7.1 Original window score and the marginal g_k term

The original joint-window energy decomposed into within-time features and cross-time interactions. Ablating the marginal score term g_k tested whether more flexible within-state score capacity explained graph recovery. It did not: gains were small and shrank with dimension.

Nodes	Pairs	Full g_k AUROC	No- g AUROC	Δ AUROC	Δ AUPRC
10	32	0.745	0.728	0.017	0.028
50	32	0.545	0.536	0.008	0.008

Table 10: Marginal-term ablation in the original SBTG window model.

Nodes	Synthetic family	Full g_k AUROC	No- g AUROC	Δ AUROC
10	Tanh mean dynamics	0.974	0.950	0.024
10	Variance dynamics	0.763	0.742	0.021
50	Tanh mean dynamics	0.625	0.609	0.016
50	Variance dynamics	0.504	0.498	0.006

Table 11: The marginal term helped slightly but did not solve high-dimensional or variance-channel recovery.

7.2 Edge-local nonlinear window energies

The next repair replaced or augmented the global feature-bilinear interaction with edge-local nonlinear bases and optional group sparsity. Improvements over FeatureBilinear were reproducible but modest, and classical conditional-mean methods remained much stronger.

Nodes	Candidate versus FeatureBilinear	Δ AUROC	95% interval	Δ AUPRC	Δ F1
50	NonlinearEdgeSparse	0.063	[0.034, 0.094]	0.093	0.088
50	HybridEdgeSparse	0.062	[0.037, 0.089]	0.090	0.084
50	NonlinearEdge	0.061	[0.031, 0.092]	0.090	0.086
100	HybridEdgeSparse	0.052	[0.033, 0.072]	0.045	0.057
100	NonlinearEdgeSparse	0.050	[0.030, 0.071]	0.047	0.058
100	NonlinearEdge	0.049	[0.029, 0.069]	0.045	0.056

Table 12: Nonlinear edge-local extensions. Improved fraction was 0.75–0.94 across the listed cells.

Nodes	Method	AUROC	AUPRC	Top-true F1
50	VAR-Ridge	0.910 ± 0.011	0.766 ± 0.024	0.703 ± 0.027
50	VAR-LASSO	0.889 ± 0.013	0.729 ± 0.019	0.673 ± 0.022
50	Polynomial SINDy-LASSO	0.864 ± 0.046	0.708 ± 0.060	0.651 ± 0.054
50	SBTG NonlinearEdgeSparse	0.665 ± 0.015	0.332 ± 0.006	0.329 ± 0.022
50	SBTG FeatureBilinear	0.578 ± 0.085	0.164 ± 0.066	0.189 ± 0.101
100	VAR-Ridge	0.876 ± 0.019	0.715 ± 0.029	0.648 ± 0.025
100	SBTG NonlinearEdgeSparse	0.588 ± 0.009	0.195 ± 0.016	0.215 ± 0.011
100	SBTG FeatureBilinear	0.499 ± 0.003	0.104 ± 0.001	0.105 ± 0.002

Table 13: Absolute graph recovery after the nonlinear SBTG repair.

7.3 Null-contrast tuning did not close the gap

Optuna searches compared selection by held-out DSM loss, temporal contrast, and a controls-first null contrast. Hyperparameter selection improved individual window models but did not overtake ridge.

Length	Selection rule	Selected method	AUROC	AUPRC
Short	Best null contrast	FeatureBilinear	0.746 ± 0.033	0.424 ± 0.027
Short	Best control	ConditionalPairDSM, temporal contrast	0.761 ± 0.038	0.460 ± 0.053
Short	Baseline	VAR-Ridge	0.882 ± 0.026	0.699 ± 0.045
Long	Best null contrast	ConditionalPairDSM	0.771 ± 0.076	0.502 ± 0.114
Long	Best control	ConditionalPairDSM, DSM loss	0.833 ± 0.027	0.608 ± 0.020
Long	Baseline	VAR-Ridge	0.898 ± 0.016	0.738 ± 0.034

Table 14: Window-model tuning: density-objective selection and graph recovery remained misaligned.

7.4 Direct conditional-score Jacobian and typed mechanism screen

Theoretically, fitting the future-only conditional score removed marginal-history contamination from the original joint score. Empirically, it improved mean-channel parent recovery but did not expose variance parents.

Measurement noise	Method	AUROC	AUPRC
0.00	ConditionalScoreJacobian, fixed	0.692 ± 0.046	0.515 ± 0.080
0.00	ConditionalScoreJacobian, stability tuned	0.710 ± 0.039	0.558 ± 0.072
0.00	VAR(10)-LASSO	0.809 ± 0.056	0.689 ± 0.113
0.00	OracleLibrary SINDy(10)	0.895 ± 0.028	0.852 ± 0.037
0.35	ConditionalScoreJacobian, fixed	0.565 ± 0.020	0.324 ± 0.036
0.35	ConditionalScoreJacobian, stability tuned	0.515 ± 0.017	0.246 ± 0.018
0.35	VAR(10)-LASSO	0.664 ± 0.022	0.460 ± 0.022
0.35	OracleLibrary SINDy(10)	0.565 ± 0.025	0.308 ± 0.049

Table 15: Conditional-score graph recovery in nonlinear gated systems.

Method variant	Truth/score channel	AUROC	AUPRC
ConditionalScoreJacobian, fixed	Mean/mean	0.863 ± 0.060	0.671
DSM-loss tuned	Mean/mean	0.732 ± 0.165	0.485
Stability tuned	Mean/mean	0.791 ± 0.040	0.530
ConditionalScoreJacobian, fixed	Variance/variance	0.501 ± 0.041	0.108
DSM-loss tuned	Variance/variance	0.511 ± 0.031	0.112
Stability tuned	Variance/variance	0.503 ± 0.032	0.107

Table 16: Typed screen: the conditional score found mean parents but not reliability/variance parents.

7.5 Gradient-readout robustness ruled out a differentiation bug

Across linear VAR, nonlinear additive, and nonlinear gated $p = 10$ systems, automatic differentiation and central finite differences had relative L2 discrepancy $1.41\text{--}1.49 \times 10^{-4}$ with Pearson/Spearman and top- k overlap equal to 1.000. Yet support AUROC remained 0.446–0.459 and Jacobian correlations were only 0.060–0.089. SmoothGrad, finite-difference readout, jacrev, Jacobian-consistency regularization, and the small oracle-Sobolev diagnostic produced no material (> 0.03 AUROC) repair.

Family	Training	AD/FD relative L2	Support AUROC	Jacobian Pearson
Linear VAR $p = 10$	None	0.000141	0.453	0.076
Linear VAR $p = 10$	Consistency	0.000143	0.457	0.083
Nonlinear additive $p = 10$	None	0.000142	0.453	0.082
Nonlinear additive $p = 10$	Consistency	0.000141	0.456	0.089
Nonlinear gated $p = 10$	None	0.000146	0.446	0.062
Nonlinear gated $p = 10$	Consistency	0.000149	0.459	0.060

Table 17: The derivative implementation was accurate for the learned network; the learned derivative was not the graph.

7.6 Residual temporal-Jacobian: the strongest SBTG repair

The residual repair constrained the score through a global temporal conditional-mean map $\mu_\theta(H)$, then read edges from $\partial\mu_\theta/\partial H$. The full paired experiment used five regenerated DGP seeds, two training seeds, 96 tuning trials, 8 selected configurations, and 2,640 completed metric rows.

Noise	Method	AUROC	AUPRC	Top-true F1
0.00	OracleLibrary SINDy(10)	0.848 ± 0.037	0.779 ± 0.055	0.668 ± 0.061
0.00	VAR(10)-LASSO	0.765 ± 0.051	0.634 ± 0.095	0.563 ± 0.067
0.00	VAR(10)-Ridge	0.717 ± 0.062	0.569 ± 0.105	0.501 ± 0.075
0.00	Polynomial SINDy	0.646 ± 0.025	0.438 ± 0.057	0.376 ± 0.048
0.00	ResidualScoreMeanDSM, fixed	0.690 ± 0.036	0.513 ± 0.070	0.453 ± 0.043
0.00	ResidualScoreMeanDSM+MSE, DSM-loss tuned	0.720 ± 0.045	0.570 ± 0.081	0.502 ± 0.055
0.00	Unconstrained score Jacobian	0.499 ± 0.015	0.228 ± 0.011	0.229 ± 0.017
0.35	VAR(10)-LASSO	0.617 ± 0.019	0.401 ± 0.031	0.366 ± 0.025
0.35	VAR(10)-Ridge	0.603 ± 0.009	0.373 ± 0.024	0.348 ± 0.009
0.35	ResidualScoreMeanDSM, null-contrast tuned	0.578 ± 0.014	0.329 ± 0.030	0.309 ± 0.019
0.35	ResidualScoreMeanDSM, fixed	0.569 ± 0.022	0.326 ± 0.039	0.312 ± 0.028
0.35	Unconstrained score Jacobian	0.496 ± 0.015	0.227 ± 0.012	0.228 ± 0.020

Table 18: Residual-score temporal-Jacobian benchmark. Bold denotes the best SBTG-family row, not the overall winner.

Interpretation. This repair explains why the “simple conditional mean” story is central. Forcing the score through $\mu_\theta(H)$ recovered much of the useful mean-channel graph signal. It did not establish that the full conditional score, variance channel, or physical mechanism had been learned.

8 Matched conditional-law and physical-field adjudication

8.1 Training details

Ridge penalties were tuned over $\{0.1, 1, 10\}$ for ridge VAR. Student- t location/scale used mean ridge 1, variance ridge 10, and degrees of freedom 3.5 or 10. Penalized Gaussian NLL used ridge 0.1 and 300 optimizer iterations. Quadratic SID used ridge 0.01; DSM variants used four corruptions. Neural Gaussian and MDN models used two hidden layers of width 64, batch size 256, at most 60 epochs, patience 9, and weight-decay/dropout grids selected on validation data. Reported results average five independent DGP seeds; lower is better.

8.2 Predictive law on matched-tail data

Method	Held-out NLL	Tail-probability RMSE
MDN $K = 3$	-0.614	0.0095
Student- t ridge	-0.530	0.0204
Ridge Gaussian	-0.294	0.0287
Gaussian NLL	-0.288	0.0250
Gaussian MLP NLL	-0.286	0.0268
Gaussian MLP DSM, $\sigma = .25$	-0.280	0.0263
Gaussian MLP DSM, $\sigma = .50$	-0.283	0.0264
Quadratic DSM, $\sigma = .75$	-0.283	0.0338
Quadratic DSM, $\sigma = 1.0$	-0.283	0.0325
Quadratic Hyvärinen	9.176	0.0385

Table 19: Matched-tail predictive performance, five DGP seeds.

8.3 Physical additive-mean field

Method	Mean-field nISE
Penalized Gaussian NLL	0.369
Ridge Gaussian	0.395
Student- t ridge	0.395
Quadratic Hyvärinen SID	0.473
Quadratic DSM SID, best	0.475
Gaussian MLP NLL	0.716
MDN $K = 3$	0.794
Gaussian MLP DSM, best	0.824

Table 20: Mean-field recovery. The earlier frozen mechanism panel also gave SINDy 0.194, sparse transition 0.201, Gaussian NLL 0.246, and Hyvärinen SID 0.268.

8.4 Physical stochastic-dispersion field

Method	Log-variance-field nISE
Penalized Gaussian NLL	0.586
Gaussian MLP DSM, $\sigma = .25$	0.789
Quadratic Hyvärinen SID	0.791
Gaussian MLP DSM, $\sigma = .50$	0.794
Gaussian MLP NLL	0.809
MDN $K = 3$	0.872
Constant-variance ridge	1.000
Quadratic DSM SID, best	1.037
Student- t ridge	2.091

Table 21: Dispersion-field recovery. Neural DSM improved slightly over neural NLL but both lost to the simpler parameterized likelihood.

8.5 Predicting a variable law did not guarantee its derivative

All Gaussian families emitted a constant standardized shape-tail probability and therefore sat at the zero-field nISE baseline 1. The MDN predicted the tail probability best but its physical shape-tail derivative had nISE 2.0165. This is

the cleanest demonstration that learning $P(Y | H)$ well in a predictive score does not guarantee accurate $D_h P$.

Method	Shape-tail field nISE	Interpretation
Gaussian/Student- t /quadratic SID families	1.000	Zero-field baseline
Gaussian MLP NLL/DSM	1.000	No fitted shape variation
MDN $K = 3$	2.017	Predictive tail win, derivative worse than zero

Table 22: Physical standardized shape-tail derivative.

8.6 Synaptic gating remained unsolved

Method	Gating-field nISE	Interpretation
Gaussian NLL / ridge / Student- t	1.000	Emit zero field; best result
Quadratic DSM SID	2.10–2.95	Worse than zero
Quadratic Hyvärinen SID	4.42	Worse than zero
Neural NLL / DSM / MDN	20.97–23.40	Highly unstable mixed derivative

Table 23: Total modulator-gated response field.

9 Mean-visible, mean-blind, and joint synthetic circuits

9.1 Earlier per-target battery

Condition	Mean R^2	Ridge	SINDy	MLP	Residual mean	Residual var	Bridge	Score Jac.	Hessian DSM
Linear	0.07	0.756	0.757	0.767	0.753	0.459	0.736	0.760	0.649
Tanh	0.03	0.776	0.766	0.777	0.763	0.469	0.745	0.763	0.637
Gated	-0.02	0.777	0.726	0.725	0.772	0.433	0.733	0.717	0.568
Branch/multimodal	-0.05	0.521	0.552	0.541	0.531	0.692	0.510	0.429	0.534
Variance gate	-0.05	0.488	0.549	0.554	0.544	0.632	0.545	0.514	0.488
Branch + noise .4	-0.05	0.493	0.517	0.501	0.569	0.606	0.497	0.422	0.521
Variance gate + noise .4	-0.05	0.497	0.509	0.526	0.474	0.541	0.527	0.488	0.476

Table 24: Graph AUROC in the earlier per-target distributional-versus-mean battery. Bold identifies the targeted second-moment win.

Mean methods win when the generator encodes parents in $\mathbb{E}[Y | H]$. The residual log-variance derivative wins only in mean-blind branch and reliability-routing cases. At measurement noise 0.4 even that advantage is fragile. The result supports a “moment ladder,” not a universal distributional estimator.

9.2 Later moment-ladder and joint-covariance adjudication

The follow-up put a plain regression and a score/diffusion reader on the same conditional-moment rung. Mean edges were read by ridge versus residual-mean or a transport bridge; diagonal variance edges by squared-residual regression versus residual-variance; and off-diagonal covariance edges by residual cross-product regression versus residual-covariance. All comparisons used ten seeds on the common $N = 40, M = 12, L = 3, T = 20,000$ backbone.

Rung	Mechanisms	Regression	Score/diffusion	Reg.	Diff.	Δ	Wilcoxon p
Mean	Linear, tanh, gated	Ridge	Residual mean	0.820	0.799	+0.020	0.020
Mean	Linear, tanh, gated	Ridge	SB bridge	0.820	0.812	+0.008	0.271
Variance	Branch, variance gate	Variance regression	Residual variance	0.678	0.635	+0.043	0.009
Covariance	Cross covariance	Cross-product regression	Residual covariance	0.698	0.636	+0.062	0.049

Table 25: Ten-seed no-free-lunch adjudication. Δ = regression minus score/diffusion AUROC. A matched regression was never worse on the conditional moment that carried the edge.

Mechanism	Mean R^2	Ridge	SINDy	MLP-J	Res.-mean	Res.-var	Res.-cov	SB bridge	Score-J	Hessian
Linear	0.11	0.807	0.798	0.802	0.791	0.487	0.392	0.811	0.791	0.648
Tanh	0.06	0.817	0.812	0.815	0.798	0.495	0.379	0.821	0.795	0.619
Gated	-0.01	0.834	0.806	0.793	0.809	0.469	0.443	0.804	0.762	0.534
Branch	-0.05	0.496	0.547	0.527	0.471	0.658	0.580	0.518	0.517	0.475
Variance gate	-0.05	0.522	0.596	0.574	0.489	0.612	0.602	0.521	0.544	0.533
Cross covariance	-0.05	0.488	0.483	0.503	0.504	0.492	0.636	0.697	0.651	0.509
Branch + noise .4	-0.05	0.492	0.511	0.494	0.500	0.582	0.522	0.534	0.486	0.448
Variance gate + noise .4	-0.05	0.504	0.549	0.566	0.480	0.557	0.580	0.514	0.509	0.526
Cross covariance + noise .4	-0.04	0.527	0.478	0.463	0.486	0.475	0.625	0.685	0.607	0.505

Table 26: Full ten-seed moment-ladder battery: off-diagonal lag-union AUROC. The table replaces the original heatmap and includes the Hessian control numerically.

Interpretation. The later result sharpened the earlier one. Distributional information can matter, but fitting a full score or transport is not necessary for graph recovery when the relevant conditional moment is declared: a regression on that moment matches or beats it. Generative models become valuable again for a different task - drawing complete, compatible path samples for finite model-relative queries.

10 Point tangents, finite contrasts, and the ORTH repairs

10.1 Point-history-gradient readiness

All 96 convergence-sweep fits completed; fixed-batch overfit checks passed; selected checkpoints did not stop at the epoch limit; exact replay reproduced the decision. Nevertheless, the preregistered derivative gates failed.

DGP	Model/readout	Tangent NRMSE	Decision
G1	Gaussian normalized model	0.589	Failed readiness threshold
G1	Autoregressive MDN	0.719	Failed readiness threshold
G2	Best clean covariance tangent	1.068	Did not beat zero
G3	Best clean skew tangent	1.167	Did not beat zero

Table 27: Optimization success did not imply derivative validity.

10.2 Ten-seed finite central contrasts

The finite panel used one clean rebuilt predictive source, $\delta = 0.1$, and ten adapter/evaluation seeds 5004–5013. All 76,760 metric rows completed. It tests adapter randomness, not ten independently trained generators.

Generator	Best practical route	Witness NRMSE	Verdict
G1 location	Signed classifier / Gaussian finite ratio	0.61–0.65	Useful
G2 covariance-only	Ratio critic	1.057	Did not beat zero
G3 skew-only	Ratio critic	1.008	Did not beat zero
G4 low-rank mixture	Ratio critic	0.656	Useful

Table 28: Finite supported contrasts. Oracle-sample classifiers were also near 1.14 NRMSE for covariance and skew, locating the bottleneck in the adapter/dictionary.

10.3 Legacy tangent-model comparison

These medians aggregate heterogeneous strata and are descriptive rather than a paired leaderboard.

Method	Metric rows	Median tangent NRMSE	Mean
Ratio critic	64	0.616	0.826
Autoregressive Transformer	16	0.629	0.694
Affine flow	64	0.946	2.098
Autoregressive MDN	64	1.015	2.030
Legacy diffusion	192	1.022	1.659
Bounded energy	64	1.181	4.396
Gaussian NLL tangent	64	1.216	4.279
Gaussian DSM	64	1.260	3.828
EDM diffusion	192	1.743	3.240
Gaussian-anchored diffusion	192	2.692	3.968

Table 29: Earlier stabilized score/generative history-tangent benchmark.

10.4 E0–E11 adjudication after targeted repairs

ID	Question	Decision	Key evidence and consequence
E0	Algebra, leakage, reproducibility	Pass	20/20 package tests and all new validation checks passed
E1	Regular equal-access recovery	Pass / no point win	ORTH NRMSE 0.104, coverage 0.940, PLUG ratio 1.002; benefit was inference, not point error
E2	Orthogonal remainder	Pass	Product-bias $R^2 > 0.99999997$
E3	One future/dependence	Pass with HAC	At $\rho = .9$, NRMSE 0.145/0.093 and HAC coverage 0.949/0.951; blocked splits/HAC required
E4	Moment-blind law change	Pass at resolving bank	Power 1.00/0.867; moment FWER 0.067/0.067; frequency bank is part of estimand
E5	Support motion	Pass	ORTH NRMSE 0.016 at zero noise where ordinary likelihood score is undefined
E6	Heavy tails/weak overlap	Conditional pass	Fixed sieve biased; density-score ORTH error 0.0036 with 1.00 coverage at gap4, $n = 128k$
E7	Heterogeneous cancellation	Pass at $R = 2$	Energy 3.982 versus oracle 4.000; local sign reversal recovered
E8	Filtered latent recovery	Conditional pass	At $\rho = .98, T = 64, n = 32k$: CF ORTH NRMSE 0.141; at $T = 4$: 2.88

ID	Question	Decision	Key evidence and consequence
E9	Dense unknown geometry	Typed pass / full-feature fail	$d_y = 8$ direct covariance NRMSE 0.106 at 128k; $d_y = 32$ low-rank 0.177; native feature NRMSE 1.01
E10	Flexible reusable laws	Local pooled pass	MDN $K = 5/10/20$ and NSF-6/10, seven families, three initializations and ten DGP seeds; no universal winner
E11	Path ordering	Pass	Full-path power 1.00; endpoint/unordered FWER 0.056

10.5 E6: fixing a representer bias floor

The original targeted random-feature Riesz sieve stopped improving with n . The repair fit a BIC-selected history density, differentiated its log density, and used cross-fitted ORTH moments. In the difficult gap- μ_4 cell, the targeted representer retained alpha NRMSE about 0.965, while the density-score alpha error decreased to 0.029.

n	Method	Target abs. error	Alpha NRMSE	Coverage
8,000	ORTH density score	0.02218	0.0659	0.90
32,000	ORTH density score	0.01180	0.0317	1.00
128,000	ORTH density score	0.00357	0.0287	1.00
128,000	ORTH targeted sieve	0.02719	0.9661	0.00
128,000	PLUG	0.03463	—	0.00

Table 31: E6 gap- μ_4 sample scaling. Severe overlap still raises the representer norm and variance.

10.6 E8: filtered latent recovery is an inverse problem

With known or accurately fitted linear filtering, exact characteristic-function correction can recover the latent target only when the observation window makes the inverse attenuation tolerable. Accurate estimation of the filter coefficient is not sufficient at short horizons.

ρ	Horizon	n	Method	NRMSE	Coverage
.98	4	8k	Exact CF ORTH, known filter	1.652	—
.98	16	8k	Exact CF ORTH, known filter	0.412	—
.98	64	8k	Exact CF ORTH, known filter	0.235	—
.98	4	32k	Exact CF ORTH, fitted mixture filter	2.880	—
.98	64	32k	Exact CF ORTH, fitted mixture filter	0.141	0.992
.98	64	32k	Latent oracle	0.049	—

Table 32: E8 horizon/measurement-model sensitivity. At $T = 64, n = 32k$, absolute ρ error was 5.5×10^{-4} .

10.7 E9: rank, inversion ridge, and typed covariance

The nominal rank-three symmetric generator had effective rank two because one population coefficient averaged to zero. An identified intercept scheme $(-1.2, -0.4, 0.9)$ restored rank three. Separately, the characteristic inverse ridge 10^{-4} caused oracle NRMSE 0.667; reducing it to 10^{-8} gave oracle NRMSE 0.0111 on the same 64-query bank.

d_y	n	Readout	Cov. NRMSE	Feature NRMSE	Coverage
8	8k	ORTH direct covariance	0.404	0.404	0.958
8	32k	ORTH direct covariance	0.219	0.219	0.943
8	128k	ORTH direct covariance	0.106	0.106	0.960
8	128k	Effective-rank projection	0.081	0.081	0.960
8	128k	ORTH characteristic inverse	0.106	1.012	0.953
32	32k	ORTH direct covariance	0.475	0.475	0.949
32	32k	Effective-rank projection	0.177	0.177	0.949

Table 33: E9 corrected identified geometry. Full characteristic-feature recovery remained much harder than the typed covariance target.

11 Changepoint experiments

11.1 Matched synthetic detection and confusion

The frozen-v2 changepoint panel used 199 matched calibration trajectories and 100 independent evaluation trajectories per condition. It separates detection from localization and checks nuisance mechanisms.

Probe	Target condition	Detection / localization	Nuisance behavior
Mean CUSUM	High-signal mean change	100/100 detected and localized	Matched channel success
Residual tail shape	Mean/variance-matched tail change	100/100 detected and localized	Matched channel success
Residual variance	Dispersion change	100/100 detected; 41% localized	Also called 95% mean and 99% tail changes; broad instability alarm
Mean / rule / variance nulls	Independent no-change	2% / 4% / 4% FPR	Near nominal
Tail null	Independent no-change	3% FPR	Near nominal
MMD / energy null	Independent no-change	9% / 11% FPR	Above nominal 5%

Table 34: Matched synthetic changepoint calibration.

11.2 June full 14-method benchmark

The last fully synthesized extended run contained 740 event windows, 10,360 method-event tests, 8,960 matched synthetic no-change calibration scans, and 14 methods. Neural p-values came from quiet pseudo-boundaries, not iid row permutations. Synthetic p-values used independent matched no-change time series.

Cohort / event	Method	Detected	Hit in 5s	MAE (s)	Median p/q	Interpretation
Imputed onset	Hotelling mean CUSUM	0.667	0.933	1.933	$q \approx .071$	Strongest marginal onset detector
Complete onset	Hotelling mean CUSUM	0.778	1.000	–	–	Consistent six-worm sensitivity
Imputed onset	Chow/ridge mean scan	0.000	1.000	0.883	.704 / .902	Localized but not significant
Imputed onset	Fixed SBTG score product	0.000	–	–	–	No calibrated hit
Imputed onset	Tuned SBTG score product	0.000	–	–	–	No calibrated hit
Imputed offset	Hotelling mean CUSUM	0.150	–	–	–	Offsets weaker than onsets
Quiet controls	All method families	–	–	–	–	Pseudo-boundary FPR: 0.050 imputed, 0.042 complete

Table 35: Primary findings from the 9 June 2026 extended changepoint benchmark.

The 14 methods were marginal CUSUM, Hotelling mean CUSUM, residualized endpoint CUSUM, residualized product CUSUM, residualized SBTG score-product CUSUM, tuned SBTG score-product CUSUM, variance CUSUM, reliability-variance CUSUM, MMD scan, energy-distance scan, local PELT L2, binary segmentation L2, window L2 scan, and Chow/ridge conditional-mean scan.

11.3 July score-dynamics follow-up

Later probes added quadratic score-dynamics mean, variance/Fisher, and joint readouts plus pooled history-DSM reliability scans. These were subsequent focused runs, not replacements for the full 14-method matrix.

Cohort	Method	n	Detected within 5s	Hit within 5s	MAE (s)	Result
Imputed butanone onset	Pooled history-DSM v2	20	0.350	–	–	Strongest calibrated conditional/reliability row
Imputed butanone onset	Score-dynamics variance	20	0.150	0.850	2.363	Good localization, weak calibrated power
Imputed butanone onset	Hotelling mean	20	0.300	–	–	Strong marginal comparator
Complete butanone onset	Pooled history-DSM v2	6	0.167	0.667	4.750	Incomplete sensitivity
Complete butanone onset	Score-dynamics variance	6	0.000	0.833	2.875	Localized, no calibrated detections
Complete butanone onset	Hotelling mean	6	0.333	0.833	2.125	Strongest complete-cohort row

Table 36: Score-dynamics follow-up. In the synthetic gate, the variance readout localized 0.667 matched-marginal, 0.667 precision-routing, and 1.000 rule-change cases, but 0.000 nonlinear-precision cases.

12 Real neural-data evidence

12.1 Cohorts and grouping

Dataset/cohort	Size	Use	Independence/claim boundary
Six-worm clean subset	6 worms	Original deconvolved lag and leave-one-worm analyses	Row splits are invalid for confirmation; group by worm
ACMMA head cohort	28 worms	Incomplete multi-worm estimator and real predictive likelihood	Animal is the inferential unit; no imputed biological edges
Corrected OH16230	17 worms, 54 classes, 4 Hz	Conditional-flow repaired-path atlas	Five held-out-worm folds; binary any-stimulus input
Historical SBTG80	20 released traces, 80 classes	Historical reproduction/context	Invalid head/tail index fusion and donor trace copying
DANDI 000981 planned	22 hermaphrodites + 22 males; expected ~ 153 shared classes	Prospective sex-aware validation	Prepared campaign, not completed evidence

Table 37: Neural cohorts used across the repository.

12.2 The original biological signal attenuated under valid grouping

Analysis	Scope	Result	Interpretation
Original peptide gain-slow	Six worms, 80% row split	DBC +3.35	Apparent strong lead, optimistic dependence structure
Grouped reanalysis	Same six worms, rows grouped by worm	DBC +1.16	Substantial attenuation
Validated all-animal ACMMA	28 worms	DBC −0.41; monoamine/peptide intervals include zero	Original lag claim did not replicate
Global-mode removal / DSM / variance-VAR	Multiple reanalyses	No stable recovery	Not rescued by objective or preprocessing variants

Table 38: Chronology of the original lag-resolved neuromodulator result.

12.3 Eight real-data levers

Eight controls-first analyses used 200 surrogate/bootstrap replicates and 5,000 permutations where applicable, on the six-worm and 28-worm configurations.

Lever	Implementation and result	Verdict
Predictive likelihood	Leave-one-worm-out variance model versus the same fitted mean with constant variance: $\Delta\text{NLL} +0.038$ on 6 worms (85% targets; CI [.033,.042]) and $+0.364$ on 28 worms (100%; CI [.245,.484])	Strongest real result; predictive, not receptor-specific

Lever	Implementation and result	Verdict
Self-gain diagonal	Exposed previously zeroed $d \log \text{Var}(Y_j)/dx_j$; initial peptide slow gain $+0.032$, $p \approx .09$	Readout valid synthetically, biological lead retracted after reliability/mechanism tests
Denoising ladder	$\sigma = 0, .25, .5, 1.0$, each variance-corrected and averaged; split-half stability 0.234 to 0.254	Modest reliability gain; weak biology
Ganglion aggregation	Pool about 8–14 ganglia; stability 0.25 to 0.54; corrected reference gains $+0.14/+0.17$, $p = .005$	Real coarsening benefit, not distributional-specific
Functional atlas	Randi/Leifer mask, 309 significant edges over 1,900 pairs; gain AUROC .555 ($p = .025$), Pearson .621	Weak static signal; simple correlation wins
Functional kinetics	Preferred gain lag versus measured atlas time scale; Spearman .015, $p = .30$	Negative
Frozen features	Freeze neural trunk, fit analytic quadratic-score head, map through trunk Jacobian; .513 versus end-to-end .503 versus raw .497	Near chance; data-limited
Global-state gating	PC1 as modulator; peptide state gain $+0.08$, $p \approx .11$; ridge sign changed $+0.23$ to -0.007	Fragile negative
Hierarchical shrinkage	James–Stein toward ganglion blocks; stability .258 to .187 at weight .5	Clean negative

12.4 Self-gain falsification

Synthetic injection passed through the actual calcium, OASIS deconvolution, and standardization pipeline. Clean recovery had Spearman 0.86 and AUROC 1.00; after calcium processing it retained Spearman 0.75 and AUROC 0.94 with a $+0.026$ group difference. At zero true effect, high variance, autocorrelation, and slower calcium produced differences $-0.010, -0.010, -0.001$. The readout was therefore valid, but the biological pattern lacked reliability and a metabotropic fingerprint.

Worms	Observation noise	Synthetic reliability	Median p	Power at .05
6	0.3	0.21	0.017	0.67
6	1.5	0.13	0.047	0.53
6	3.0	0.08	0.250	0.33
28	3.0	0.47	0.067	0.47

Table 40: Synthetic self-gain power after degradation. Real cross-estimator per-neuron reliability was about 0.05.

Partial correlation test	Six worms	28 worms
Slow self-gain vs. metabotropic receptor load	-0.13 ($p = .28$)	-0.03 ($p = .79$)
Slow self-gain vs. ionotropic load	-0.09	-0.11
Double dissociation, metabotropic minus ionotropic	-0.04	$+0.07$
Slow self-gain vs. neuropeptide GPCR	$+0.03$ ($p = .77$)	$+0.27$ ($p = .02$)

Table 41: The core metabotropic prediction failed; the single 28-worm peptide-GPCR flicker did not replicate in six worms.

13 Compatibility-aware response sampling

13.1 Corrected 54-neuron implementation

Setting	Value	Role
Observed state	54 standardized neurons, 17 OH16230 worms, native 4 Hz	Corrected head-only cohort
History	80 frames / 20 s	Finite-memory conditional transition input
Source window	4 frames / 1 s	Average activity used by the clamp
Explicit lags	1, 4, 8, 16 frames	0.25, 1, 2, 4 s from source-window end to cut
Source targets	Training-fold Q25/Q75	Phase-specific low/high states
Bandwidth	$\max(.25 \text{ IQR}, .10)$	Soft-clamp bias/overlap tradeoff
Population anchor	Rank-12 training-fold PCA, $\lambda = .25$	Keeps non-source context compatible
Particle screen	32 survivors; progressive branch 2, future factor 2	Frozen E26 comparison
Bridge ESS floor	65% of candidates	Adaptive tempering increment
Future channels	Endpoint/cumulative mean, peak, event probability, SD, log-SD, W1	Compares future laws, not means alone
Orientation	Target rows $\times$ source columns	External comparison convention

Table 42: Frozen repaired-path configuration.

13.2 Four samplers

Workflow	Natural bank	Sequential	Bridge	Source resampling	Main tradeoff
Direct importance	Yes	No	No	Never	Fast; rare queries collapse
Terminal SMC	No	Yes	No	Deferred to cut	Equal-weight future particles; terminal collapse
Progressive bridge	No	Branched	Across source window	ESS-adaptive	Lowest sealed particle error; expensive
Temporal cut	No	Yes	No	From source-window end	Explicit historical placement; early ancestry loss

Table 43: Sampler taxonomy. The first three mainly differ as estimators; temporal cut also changes constraint placement.

13.3 Sealed Stage-A estimator benchmark

The comparison used one frozen flow checkpoint, two independent 4,096-particle references, and three finite-particle repeats. No atlas was used for selection.

Estimator	MSE	Bias ²	MC variance	Signed ρ	Valid rate	Runtime (min)
Direct importance	0.030689	0.013059	0.017630	0.758	0.440	0.5
Terminal SMC	0.015645	0.008098	0.007547	0.815	0.435	21.2
Progressive bridge SMC	0.011008	0.006580	0.004428	0.854	0.911	34.3

Table 44: Finite-particle error against the same learned repaired law, not biological truth.

13.4 External correspondence on the corrected 54-neuron axis

Method	Randi	Cook structural	Cook chemical	Cook gap
Progressive bridge N=32	0.647	0.611	0.600	0.672
Direct importance N=256	0.617	0.619	0.610	0.660
Published SBTG, restricted to 54	0.622	0.565	0.558	0.621
Current SBTG	0.524	0.524	0.517	0.585

Table 45: Lag-1/horizon-1 AUROC on identical 54-class denominators. External reference correspondence is contextual, not causal validation.

Continuous agreement was weaker: progressive Randi absolute-response Spearman was 0.211; Cook structural Spearman was 0.172 over all pairs and 0.064 among positive edges. The corrected complete-family analysis retained 102 baseline-mean edges and 492 cells after joint max- T , but zero strong-support non-flat lag profiles and zero strict stimulus-modulated discoveries. In the targeted N=128 calibration, zero rows exceeded both sampler and quiet-time controls.

14 Model error and out-of-distribution queries

14.1 Predictive qualification across six families

Model	Energy	CRPS	RMSE	90% coverage	Tail Brier
EDM diffusion	1.0371	0.1005	0.1985	0.9007	0.06385
Flow	1.0454	0.1022	0.1998	0.8917	0.06964
MDN-4	1.0788	0.1018	0.2056	0.8501	0.06149
Gaussian rank 8	1.0878	0.1034	0.2069	0.8843	0.06165
Student- t rank 8	1.0879	0.1030	0.2070	0.8927	0.06135
Diagonal Gaussian	1.0888	0.1033	0.2072	0.8743	0.06163

Table 46: All six model families passed the study’s predictive qualification gate.

The synthetic suite included ten seeds spanning nonlinear location, state-dependent variance/covariance, heavy tails, switching, multimodality, hidden global state, calcium filtering, and independently parameterized eight-node FitzHugh–Nagumo networks with common-random-number pulse truth.

Finding	Value	Status	Meaning
Supported-query between-family SD	0.01077	Validated	Families broadly agree on in-support queries
Rare query, direct matched cost	MAE 0.1189	Observed	Beat progressive under matched compute
Rare query, progressive bridge	MAE 0.1559	Observed	Beat same- N direct, not matched-cost direct
Rare query, direct same N	MAE 0.2960	Observed	Severe finite-particle error
Error vs. minimum ESS	Spearman -0.509	Validated	ESS informative but incomplete
Error vs. history distance	Spearman 0.126	Validated	This scalar distance was weak
Error vs. cross-model disagreement	Spearman 0.797	Validated	Strongest tested warning signal
Stable but wrong	16 rows	Validated	Sampler gates passed while learned-model error exceeded 0.25

Table 47: Particle error and learned-model error are separate.

Interpretation. Progressive SMC can repair a particle approximation to a fixed model-relative query. It cannot repair extrapolation bias, omitted state, a wrong conditional law, or a biologically invalid query. Every reported response therefore needs event rarity, empirical support, cross-model spread, and particle convergence together.

15 Historical 80-class atlas and current program

The completed historical atlas contains 20 raw archives across four lags, 13,977,600 dense response cells, seven channels, thirteen contexts, six horizons, 80-by-80 matrices, and 256 whole-worm bootstrap replicates. It preserves the released SBTG80 lineage for comparison but cannot become prospective evidence because that cache fused head and tail recordings by index and copied donor traces into missing classes.

Method	Reference	AUROC	Source-macro AUROC
Progressive bridge	Randi wild type	0.657	0.623
Progressive bridge	Cook structural 80	0.614	0.601
Progressive bridge	Cook chemical 80	0.605	0.587
Progressive bridge	Cook gap 80	0.654	0.680
Published SBTG	Randi wild type	0.630	0.615
Published SBTG	Cook structural 80	0.581	0.578
Published SBTG	Cook chemical 80	0.570	0.557
Published SBTG	Cook gap 80	0.613	0.652

Table 48: Historical/contextual lag-1 correspondence. The data lineage prevents prospective interpretation.

Artifact	State	Scope	Claim boundary
Historical 80-class atlas	Complete	20 traces, 80 classes, four lags, six horizons, seven channels	Context only because of invalid fusion/imputation lineage
Corrected 54-neuron atlas	Complete negative lag adjudication	17 native-4-Hz worms with onset and matched quiet controls	No experiment-ready lag edge passed both gates
OOD robustness suite	Complete	Six model families, nonlinear VAR, FHN, NeuroPAL pseudo-OOD	ESS alone is not scientific validity
DANDI 000981 campaign	Prepared, not executed	22 hermaphrodites + 22 males, expected ~ 153 shared exact labels	Prospective observational/model-relative validation

Table 49: Repository status at the report freeze.

15.1 Prepared DANDI 000981 campaign

Design element	Frozen plan	Reason
Cohort	Exactly 22 retained hermaphrodites and 22 retained males	Stop if runtime exclusions differ
Neuron axis	Exact-name intersection observed in at least 20/22 subjects per sex; minimum 101, expected about 153	Same ordered axis across all models
Models	Combined shared90, hermaphrodite shared90, male shared90	Combined model alone selects preprocessing/architecture
Cross-validation	Five whole-subject folds; validation fold $(k + 1) \bmod 5$	No window-level leakage
Preprocessing screen	Raw, causal MA3, causal MA11–MA241, causal 90-s EWMA; noncausal published filter only sensitivity	Causal primary pipeline
Sampling	Direct, terminal, progressive, temporal-cut	Same four estimators in every analysis
Lags	1,2,3,5,11,21,32,53 frames at 2.66 Hz	About 0.376–19.925 s
Horizons	1,2,4,8,16 frames; timing jitter $-1, 0, +1$	Timing robustness
Primary estimand	Onset minus matched-quiet endpoint-mean response	Event specificity before external atlas use
External firewall	Cook/Randi/Bentley/SBTG loaded only after matrices and sampler selection freeze	Avoid atlas-driven tuning

Table 50: Prospective campaign design.

Stage	Combined	Hermaphrodite	Male	Total
NWB extraction	47 shared	–	–	47
Preprocessing screen	50	–	–	50
Encoding screen	70	–	–	70
History screen	60	–	–	60
Architecture/density screen	160	–	–	160
Three-seed confirmation	15	15	15	45
Four-method primary sampling	3,840	3,840	3,840	11,520
Particle convergence	1,920	1,920	1,920	5,760
All array tasks	–	–	–	17,712

Table 51: Prepared DANDI task inventory. These are planned jobs, not completed results.

16 What the complete program now supports

Conclusion	Status	Evidence boundary
Simple mean models are the default for mean dynamics	Supported	Ridge/VAR/SINDy repeatedly win mean-visible simulations and explain the residual-score SBTG repair
Conditional variability matters for prediction	Supported	Real held-out $\Delta\text{NLL} +0.364$ on 28 worms; synthetic second-moment wins in mean-blind cases
A good predictive law guarantees a good history derivative	Rejected	MDN tail prediction versus wrong tail derivative; tangent failures despite good NLL/score fit

Conclusion	Status	Evidence boundary
Original SBTG identifies physical neuromodulator edges	Rejected	Weak mechanism specificity, near-chance variance parents, nonreplicated biological lag signature
Finite supported contrasts can recover selected law changes	Supported, scoped	G1/G4 finite witnesses, G8 path feature, repaired E6/E9 typed targets
One generic finite classifier handles covariance and skew	Rejected	Oracle-sample adapters failed G2/G3 at the original resolution
Progressive bridge SMC improves finite-particle estimation	Supported	Stage-A MSE/variance/validity versus the same learned-law target
Progressive SMC validates an unsupported model query	Rejected	Sixteen stable-but-wrong rows; disagreement correlated 0.797 with total error
Current neural matrices are causal/anatomical graphs	Rejected	Observational fitted-law contrasts and external correspondence only
The next campaign can test transfer prospectively	Prepared	DANDI protocol is frozen; execution and results remain outstanding

17 Reproducibility and evidence map

Evidence lane	Repository artifact
Matched objective results	SYNTHETIC_METHODS_REPORT_2026-07-12.md; neuromod_benchmark/outputs/objective_adjudication_20260712/
Real-data synthesis	EXPERIMENT_REVIEW_2026-07-12.md
Eight neural levers	sid_elegans/output/newlevers/RESULTS.md and VALIDATION.md
History-tangent DGPs	history_tangent_benchmark/src/history_tangent_benchmark/dgps.py
Finite contrast protocol	history_tangent_benchmark/REVISED_NOTE_V1_AUDIT.md
Diffusion/autoregression audit	history_tangent_benchmark/DIFFUSION_VS_AUTOREGRESSION_AUDIT.md
ORTH adjudication	reports/distributional_sid_adjudication_2026-07-14/
SBTG synthetic chronology	/Users/vik/Developer/diffusionCircuit/docs/sbtg_synthetic_experiment_synthesis.md
Moment-ladder/covariance battery	/Users/vik/Developer/diffusionCircuit/docs/distributional_circuit_recovery.tex; /Users/vik/Developer/diffusionCircuit/docs/nofreelunch_circuit_recovery.tex
Changepoint results	/Users/vik/Developer/diffusionCircuit/docs/RESULTS.md and dated benchmark folders
Repaired-path specification	FLOW_REPAIRED_LAG_METHODS_20260828.md
Compatibility experiment registry	results/compatibility_path_response/EXPERIMENT_INDEX.md
Corrected 54-neuron atlas	results/neural_prediction_atlas_20260829/README.md
Neuron-class interpretation	results/neuron_class_effects_20260831/README.md
OOD/model-error study	results/query_ood_robustness_20260901/REPORT.md
Historical 80-class atlas	results/sbtg80_optimized_full_atlas_20260902/REPORT.md
Prospective DANDI campaign	dandi000981_lag_cluster_bundle/README_CLUSTER.md

All aggregate values in the report were transcribed from reviewed local reports, machine-readable summaries, manifests,

or current code defaults. Raw row-level exports remain in the listed result folders. The report does not reproduce millions of row-level records; it includes the complete set of key protocol parameters and decision-relevant aggregate results needed to audit the narrative.

18 Conclusion

The program did not discover that “SBTG simply needs a larger neural network.” It discovered a sequence of estimand distinctions. Joint-window score products could localize some mean-like dependence but were weak for the distributional mechanisms that motivated them. Conditional-score cleanup and derivative regularization ruled out several coding and numerical explanations. The strongest SBTG repair forced the score through a conditional-mean transition map, which made explicit that the recovered signal was primarily mean dynamics. Simple mean models consequently remained the strongest default.

Synthetic systems then showed exactly where this default ends. Covariance-only, skew-only, multimodal, reliability, and higher-order path changes can leave the conditional mean unchanged. Predictive densities and samplers can represent those changes, but downstream history derivatives can remain unstable, and dataset-level averaging can cancel heterogeneous effects. The viable alternative is not an unconstrained point derivative. It is a finite, supported, typed contrast whose response feature is declared before fitting and whose adapter passes an oracle gate.

Compatibility-aware repaired paths instantiate that idea for neural trajectories. A source history is softly constrained inside a complete generated population context, the constraint is released at a declared cut, and the subsequent learned dynamics generate the future law of interest. Progressive bridge SMC is currently the most accurate finite-particle estimator of this frozen learned-law target. The final scientific claim remains deliberately narrow: the result is a model-implied predictive response under compatible histories. It becomes useful only when predictive qualification, empirical support, cross-model agreement, particle diagnostics, quiet-time specificity, and prospective validation agree.

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